2(a)(i)	$(\Delta)p = \rho g(\Delta)h$ $520 = 1000 \times 9.81 \times h$	**C1**
	$h = 0.053$ m	**A1**
2(a)(ii)	(upthrust =) $(\Delta)p \times A$ $= (\Delta)p \times \pi(d/2)^2$ **or** $(\Delta)p \times \pi r^2$	**C1**
	$= 520 \times \pi(0.031/2)^2 = 0.39$ (N)	**A1**
2(a)(iii)	$T = 0.84 - 0.39$ $= 0.45$ N	**A1**

Question	Answer	Marks
2(b)(i)	$a = (v - u) / t$ **or** $(\Delta)v / (\Delta)t$ **or** gradient	C1
	$= \text{e.g. } 8.0 \times 10^{-2} / 2.0$ $= 4.0 \times 10^{-2}\,\text{m}\,\text{s}^{-2}$	A1
2(b)(ii)	distance = $(½ \times 2.5 \times 0.10) + (½ \times 1.5 \times 0.10)$ **or** $(½ \times 4.0 \times 0.10)$ (= 0.20 (m))	C1
	depth = 0.32 – 0.20 = 0.12 m	A1
2(c)(i)	viscous (force)	B1
2(c)(ii)	viscous force increases (with speed/time/depth)	B1
	(so) acceleration decreases	B1